(a) m = n
(b) m > n
(c) m < n
(d) Cannot be determined to due to insufficient data

Ans. (a) : Quantity of milk in X = 100 ml
Quantity of water in Y = 100 ml
After first process–
Total number of mixture in container = 120 ml
In container ratio of water and milk = 1 : 5
After second process–
Quantity of milk in container X = 80 + 3.33 = 83.33 ml
Ratio of water in container Y = $100 \times \frac{5}{6} = 83.33$ ml
Hence, m = n

70. A pie chart gives the expenditure on five different items A, B, C, D and E in a household. If B, C, D and E correspond to 90°, 50° 45° and 75° respectively, then what is the percentage of expenditure on item A ?

(a) $\frac{112}{9}$ (b) $\frac{125}{6}$
(c) $\frac{155}{9}$ (d) $\frac{250}{9}$

Ans. (d) : Central angle for item 'A'.
= 360° – (90° + 50° 45° + 75°)
= 360° – 260° = 100°
Percentage of expenditure on item
$= \frac{100}{360} \times 100 = \frac{250}{9}\%$

Directions for the following 3 (three) items
Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage –1

To encourage research is one of the functions of a university. Contemporary universities have encouraged research, not only in those cases where research is necessary, but on all sorts of entirely unprofitable subjects as well. Scientific research is probably never completely valueless. However silly and insignificant it may seem, however mechanical and unintelligent the labours of the researchers, there is always a chance that the results may be of value to the investigator of talent, who can use the facts collected for him by uninspired but industrious researchers as the basis of some fruitful generalization. But where research is not original, but consists in the mere rearrangement of existing materials, where its object is not scientific but literary or historical, then there is a risk of the whole business becoming merely futile.

71. The author's assumption about scientific research is that

(a) it is never very valuable
(b) it is sometime very valuable
(c) it is never without some value
(d) it is always very valuable

Ans. (c) : The author's assumption about scientific research is that a scientific research is never without some value.

72. According to the author

(a) not many research results can be of value to an intelligent investigator
(b) a research result is always valuable to an intelligent investigator
(c) any research result can be value to an intelligent investigator
(d) a research result must always be of some value to an intelligent investigator

Ans. (c) : According to the author, any research result can be of value to an intelligent investigation.

Passage– 2

How best can the problems of floods and droughts be addressed so that the losses are minimal and the system becomes resilient? In this context, one important point that needs to be noted is that India gets 'too much' water (about 75% of annual precipitation) during 120 days (June to September) and 'too little' for the remaining 245 days. This skewed water availability has to be managed and regulated for its consumption throughout the year.

73. Which one of the following best reflects the practical, rational and lasting solution?

(a) Constructing huge concrete storage tanks and canals across the county
(b) Changing the cropping patterns and farming practices
(c) Interlinking of river across the country
(d) Buffer stocking of water through dams and recharging aquifers

Ans. (d) : Buffer stocking of water through dams and recharging aquifers best reflects the practical, rational and lasting solution.

74. If $15\times14\times13\times....\times3\times2\times1 = 3^m \times n$ where m and n are positive integers, then what is the maximum value of m?

(a) 7
(b) 6
(c) 5
(d) 4

Ans. (b) : From question,
$15\times14\times13\times12\times11\times10\times9\times8\times7\times6\times5\times4\times3\times2\times1 = 3^m \times n$
$3^6\times5\times14\times13\times4\times11\times10\times8\times7\times2\times5\times4\times2\times1 = 3^m \times n$
$\therefore$ m = 6

75. What is the value of X in the sequence 2, 12, 36, 80, 150, X?

(a) 248 (b) 252
(c) 258 (d) 262

Ans. (b) : Given series is as follows–
2, 12, 36, 80, 150, X
$2 = 1^2 + 1^3$
$12 = 2^2 + 2^3$
$36 = 3^2 + 3^3$
$80 = 4^2 + 4^3$
$150 = 5^2 + 5^2$
$X = 6^2 + 6^3 = 36 + 216$
X = 252

76. One non-zero digit, one vowel and one consonant from English alphabet (in capital) are to be used in forming passwords, such that each password has to start with a vowel and end with a consonant. How many such passwords can be generated?

(a) 105 (b) 525
(c) 945 (d) 1050

Ans. (c) : Hence, password is one non-zero digit one vowel and one consonant formed English alphabet.
So, five vowels can be used in the first place, nine digit can be used in second place and 21 consonant can be used in the third place.
Total number of password = 5×9×21= 945

77. There are 9 cups placed on a table arranged in equal number of rows and columns out of which 6 cups contain coffee and 3 cups contain tea. In how many ways can they be arranged so that each row should contain at least one cup of coffee?

(a) 18 (b) 27
(c) 54 (d) 81

Ans. (d) : Case-I–If one cup of coffee is placed in each line
CCT, CTC, TCC
Total ways = 3 × 3 × 3 = 27
Case-II–If three cups of coffee are placed in one line and two cups of coffee in the other and one cup of coffee in the third, then
CCC, CCT or CTC or TCC, CTT or TCT or TTC
Total ways = 3! × 3 × 3 = 54
From Case-I and Case-II
Total ways = 27 + 54 = 81

78. The sum of three consecutive integers is equal to their product. How many such possibilities are there?

(a) Only one
(b) Only two
(c) Only three
(d) No such possibility is there

Ans. (c) : Let three consecutive integer x, (x + 1) and (x + 2)
$\therefore$ $x+(x+1)+(x+2) = x(x+1)(x+2)$
$3x + 3 = x(x+1)(x+2)$
$3(x+1) = x(x+1)(x+2)$
$0 = (x+1)[x^2 + 2x - 3]$
If $x + 1 = 0$
$x = -1$
If $x^2 + 2x - 3 = 0$
$(x+3)(x-1) = 0$
$\therefore$ $x = -3$ or 1
Hence, there are three possibilities.

79. What is the number of numbers of the form 0 XY, where X and Y are distinct non-zero digits?

(a) 72
(b) 81
(c) 90
(d) 100

Ans. (a) : x and y is non digit.
Numbers between 11 and 20 = 12, 13, 14, 15, 16, 17, 18, 19 = 8
Numbers between 20 and 30 = 21, 23, 24, 25, 26, 27, 28, 29 = 8
Thus, total numbers between 31 and 100 = 7 × 8 = 56
Total number = 65+ 8 + 8 = 72

80. The average weight of A, B, C, is 40 kg, the average weight of B, D, E is 42 kg and the weight of F is equal to that of B. What is the average weight of A, B, C, D, E and F?

(a) 40.5 kg
(b) 40.8 kg
(c) 41 kg
(d) Cannot be determined as data is inadequate

Ans. (c) : According to the question,
A + B + C = 40 × 3 = 120
B + D + E = 42 × 3 = 126
$\therefore$ F = B
$\therefore$ A + B + C + D + E + F = A + B + C + D + E + B
= 120 + 126 = 246
Average weight= $\frac{246}{6} = 41$ kg